

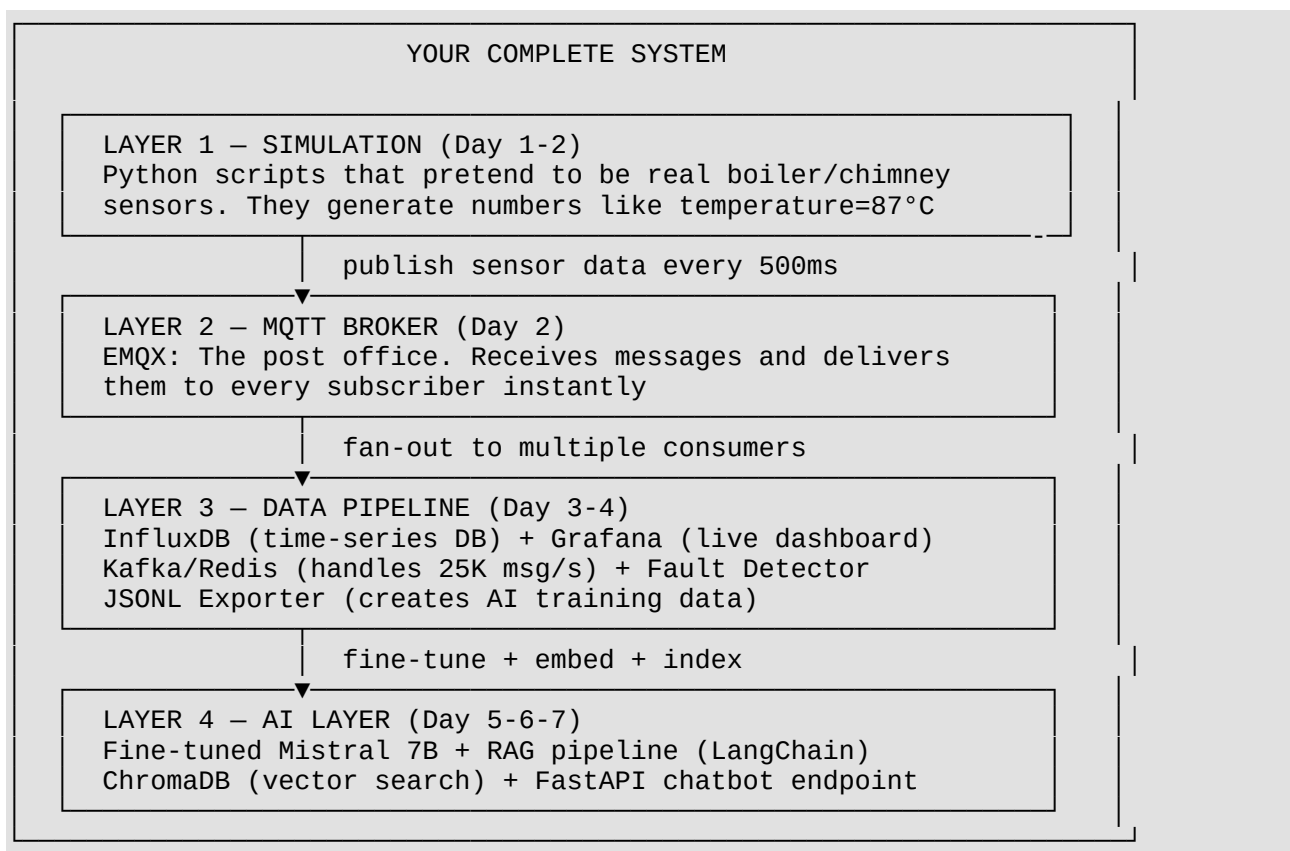
Boiler & Chimney IoT System — 7-Day Build Guide

From Zero to Real-Time AI Chatbot | Fresher-Friendly, Step-by-Step

Who this is for: You are new to IoT and MQTT. This guide assumes you know basic Python and can use a terminal. Every concept is explained before you use it. Every day has a clear goal, all the code you need, and a "what you built today" checklist.

□ System Architecture (Big Picture)

Before writing a single line of code, understand what you are building across all 7 days:



Glossary — Read This First

These are terms you will see every day. Read this once so nothing surprises you.

Term	Plain English Explanation
IoT	Internet of Things — physical devices (sensors, machines) that send data over a network
MQTT	A messaging protocol. Like WhatsApp but for machines. Lightweight, fast, made for sensors

Term	Plain English Explanation
Broker	The MQTT "post office" — receives messages from senders and delivers to receivers
Publisher	A device/script that <i>sends</i> a message to the broker (your boiler simulator)
Subscriber	A device/script that <i>receives</i> messages from the broker (your database writer)
Topic	The "address" of a message. Like boiler/temperature . Publishers and subscribers must use the same topic
QoS	Quality of Service — how reliably a message is delivered (0=fire and forget, 1=at least once, 2=exactly once)
Payload	The actual data inside a message. Example: <code>{"temperature": 87.4, "unit": "C"}</code>
Time-series DB	A database optimised for data that changes over time. Perfect for sensor readings
InfluxDB	A popular time-series database. You will store all sensor data here
Grafana	A tool that reads from InfluxDB and shows live charts on a dashboard
Kafka	A high-speed message queue. Sits between MQTT and your database for 25K+ msg/s
Docker	A tool that runs software in isolated containers. You use it to run EMQX, InfluxDB, Grafana
Fine-tuning	Teaching an existing AI model new domain knowledge using your own data
LoRA	Low Rank Adaptation — a technique to fine-tune a large LLM efficiently on a single GPU
RAG	Retrieval Augmented Generation — before answering, the AI fetches relevant real data and uses it
Vector DB	A database that stores text as mathematical vectors so you can search by meaning, not keywords
ChromaDB	A local vector database — used for RAG retrieval
Embedding	Converting text (like "boiler pressure is high") into a list of numbers that capture its meaning
LangChain	A Python library that connects LLMs, vector DBs, and data sources into one pipeline
FastAPI	A Python web framework for building APIs. You use it to expose your chatbot as an HTTP

Term	Plain English Explanation
	endpoint

7-Day Schedule Overview

Day	Focus	Output
Day 1	Environment setup + MQTT fundamentals	Docker running, EMQX broker live, first publish/subscribe working
Day 2	Boiler & chimney simulator	Python scripts generating realistic sensor data with fault injection
Day 3	InfluxDB + Grafana pipeline	All sensor data stored, live dashboard visible in browser
Day 4	Kafka buffer + JSONL exporter	25K msg/s capable, training dataset generated
Day 5	Fine-tuning the LLM	Mistral 7B fine-tuned on your boiler data using LoRA
Day 6	RAG pipeline + ChromaDB	Chatbot retrieves real-time sensor context before answering
Day 7	FastAPI chatbot + Integration	Full system working end-to-end, chatbot answers from live data

DAY 1 — Environment Setup + MQTT Fundamentals

☐ Goal

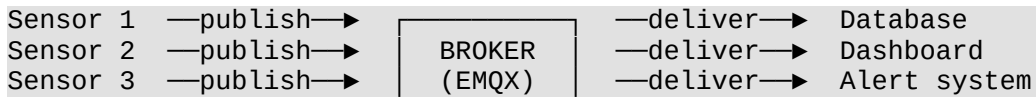
By end of Day 1 you will:

- Have Docker installed and running
- Have EMQX (MQTT broker) running in a container
- Write your first publisher and subscriber in Python
- Understand exactly what MQTT does

Concept: What is MQTT and Why Does It Exist?

Imagine you have 100 temperature sensors on a factory floor. You want all of them to send data to a central server. Without MQTT, each sensor would need to know the server's address and maintain a connection. If the server changes, you update 100 sensors.

MQTT solves this with a **broker in the middle**:



The sensors don't know about the database. The database doesn't know about the sensors. They only know about the broker. This is **decoupling** — the core benefit of pub/sub architecture.

MQTT Topic Structure

Topics are like folder paths. They use / as a separator:

boiler/temperature	← specific sensor
boiler/pressure	← another specific sensor
boiler/#	← wildcard: ALL boiler sensors
chimney/flue_gas/co2	← nested topic
system/faults	← fault events
+/temperature	← wildcard: temperature from ANY device

The # wildcard means "everything below this level." The + wildcard means "any single level."

Step 1.1 — Install Prerequisites

```
# Install Docker Desktop from https://www.docker.com/products/docker-desktop
# Verify it works:
docker --version
docker-compose --version

# Install Python 3.10+
python --version

# Create your project folder
mkdir boiler-iot-system
cd boiler-iot-system

# Create a Python virtual environment
python -m venv venv

# Activate it
# On Windows:
venv\Scripts\activate
# On Mac/Linux:
source venv/bin/activate

# Install Python MQTT library
pip install paho-mqtt
```

Step 1.2 — Run EMQX Broker with Docker

Create a file called `docker-compose.yml` in your project folder:

```
version: '3.8'

services:
  emqx:
    image: emqx/emqx:latest
    container_name: emqx_broker
    ports:
      - "1883:1883"    # MQTT port (your Python scripts connect here)
      - "8083:8083"    # MQTT over WebSocket
      - "18083:18083"  # EMQX web dashboard
    environment:
      - EMQX_ALLOW_ANONYMOUS=true # No password needed for development
    volumes:
      - emqx_data:/opt/emqx/data
    restart: unless-stopped

volumes:
  emqx_data:
```

Start it:

```
docker-compose up -d emqx
```

Open your browser: `http://localhost:18083` Login: `admin / public`

You will see the EMQX dashboard. This is your broker's control panel.

Step 1.3 — Your First Publisher (Hello World)

Create `day1_publisher.py`:

```
"""
Day 1 Publisher – sends a test message to the MQTT broker every second.
Think of this as the boiler sending one temperature reading.
"""

import paho.mqtt.client as mqtt
import json
import time

# ——— CONFIGURATION ———
BROKER_HOST = "localhost"    # EMQX is running on your machine
BROKER_PORT = 1883           # Standard MQTT port
TOPIC = "test/hello"         # The "address" of this message
CLIENT_ID = "publisher_001"  # Unique ID for this client
# ———

def on_connect(client, userdata, flags, rc):
    """Called automatically when connection to broker is established."""
    if rc == 0:
        print("▣ Connected to EMQX broker successfully!")
    else:
        print(f"▣ Connection failed with code: {rc}")

# Create an MQTT client instance
client = mqtt.Client(client_id=CLIENT_ID)
```

```

# Register the callback function
client.on_connect = on_connect

# Connect to the broker
client.connect(BROKER_HOST, BROKER_PORT, keepalive=60)

# Start network loop in background thread
client.loop_start()

# Wait for connection to establish
time.sleep(1)

print("👉 Starting to publish messages...")

message_count = 0
while True:
    message_count += 1

    # Create a JSON payload (real data would come from a sensor)
    payload = {
        "message_id": message_count,
        "text": "Hello from boiler system!",
        "timestamp": time.time()
    }

    # Convert dict to JSON string
    payload_str = json.dumps(payload)

    # Publish to broker
    # QoS=1 means "deliver at least once" (safe for sensor data)
    result = client.publish(TOPIC, payload_str, qos=1)

    print(f"👉 Published message #{message_count}: {payload_str}")

    # Wait 1 second before next message
    time.sleep(1)

```

Step 1.4 — Your First Subscriber

Create `day1_subscriber.py` (run this in a SECOND terminal):

```

"""
Day 1 Subscriber – listens for messages on a topic and prints them.
Think of this as the database consumer receiving boiler data.
"""

import paho.mqtt.client as mqtt
import json

BROKER_HOST = "localhost"
BROKER_PORT = 1883
TOPIC = "test/hello"          # Must match publisher topic
CLIENT_ID = "subscriber_001"

def on_connect(client, userdata, flags, rc):
    """Called when connected to broker."""
    if rc == 0:
        print("👉 Subscriber connected to broker!")
        # Subscribe to topic AFTER connecting

```

```

        client.subscribe(TOPIC, qos=1)
        print(f"👉 Subscribed to topic: {TOPIC}")
    else:
        print(f"❌ Connection failed: {rc}")

def on_message(client, userdata, message):
    """Called AUTOMATICALLY every time a new message arrives on subscribed
    topic."""

    # Decode bytes → string
    payload_str = message.payload.decode("utf-8")

    # Parse JSON string → Python dict
    payload = json.loads(payload_str)

    print(f"👉 Received on [{message.topic}]: {payload}")
    print(f"    Message #{payload['message_id']} arrived at timestamp
    {payload['timestamp']:.2f}")

# Setup client
client = mqtt.Client(client_id=CLIENT_ID)
client.on_connect = on_connect
client.on_message = on_message

# Connect and run forever (blocking)
client.connect(BROKER_HOST, BROKER_PORT, keepalive=60)
print("⌚ Waiting for messages... (Press Ctrl+C to stop)")
client.loop_forever() # This blocks and handles all incoming messages

```

Run both scripts

Terminal 1:

```
python day1_subscriber.py
```

Terminal 2:

```
python day1_publisher.py
```

You will see messages flowing from publisher → broker → subscriber in real time. **This is the entire foundation of your system.**

☐ Day 1 Checklist

- ☐ Docker and docker-compose installed
 - ☐ EMQX running, dashboard accessible at localhost:18083
 - ☐ Publisher script sends messages successfully
 - ☐ Subscriber receives messages in real time
 - ☐ You understand what topic, publish, subscribe, broker mean
-
-

DAY 2 — Boiler & Chimney Simulator with Fault Injection

□ Goal

By end of Day 2 you will have:

- A realistic Python boiler simulator publishing to 8+ MQTT topics
- A chimney simulator running in parallel
- A fault injector that randomly creates equipment problems
- All data in proper JSON format with timestamps

Concept: What Data Does a Real Boiler Generate?

A real industrial boiler has these sensors:

BOILER SENSORS		CHIMNEY SENSORS	
Steam Pressure	→ 10-14 bar	Flue Gas Temp	→ 150-250°C
Water Temperature	→ 60-100°C	CO2 Percentage	→ 8-14%
Fuel Flow Rate	→ 5-20 L/min	O2 Percentage	→ 3-8%
Air Flow Rate	→ 100-300 m/h	CO (Carbon Mono)	→ 0-50 ppm
Drum Water Level	→ 40-60%	Draft Pressure	→ -2 to -5 Pa
Burner State	→ ON/OFF	Stack Velocity	→ 3-8 m/s
Feed Water Flow	→ 5-20 L/min		

Normal ranges are critical — when a value goes outside the normal range, it is a fault.

Step 2.1 — Boiler Simulator

Create `simulators/boiler_simulator.py`:

```
"""
Boiler Simulator – generates realistic boiler sensor data and publishes via
MQTT.
Simulates normal operation + gradual drift + sudden faults.
"""

import paho.mqtt.client as mqtt
import json
import time
import random
import math
from datetime import datetime

# — MQTT CONFIG —
BROKER_HOST = "localhost"
BROKER_PORT = 1883
CLIENT_ID = "boiler_simulator"

# — TOPIC DEFINITIONS —
# 19 real-world parameters from a power-plant boiler + turbine.
```



```

# Boiler-side sensors go under boiler/..., turbine-side under turbine/...
TOPICS = {
    # Boiler side
    "main_steam_flow": "boiler/main_steam_flow",
    "main_steam_temp_boiler": "boiler/main_steam_temp",
    "main_steam_pressure_boiler": "boiler/main_steam_pressure",
    "reheat_steam_temp_boiler": "boiler/reheat_steam_temp",
    "superheater_desup_flow": "boiler/superheater_desup_water_flow",
    "reheater_desup_flow": "boiler/reheater_desup_water_flow",
    "feedwater_temp": "boiler/feedwater_temp",
    "feedwater_flow": "boiler/feedwater_flow",
    "feedwater_pressure": "boiler/feedwater_pressure",
    "flue_gas_temp": "boiler/flue_gas_temp",
    "oxygen_level": "boiler/oxygen_level",
    # Turbine side
    "main_steam_temp_turbine": "turbine/main_steam_temp",
    "main_steam_pressure_turbine": "turbine/main_steam_pressure",
    "reheat_steam_temp_turbine": "turbine/reheat_steam_temp",
    "reheat_steam_pressure_turbine": "turbine/reheat_steam_pressure",
    "control_stage_pressure": "turbine/control_stage_pressure",
    "high_exhaust_pressure": "turbine/high_exhaust_pressure",
    "condenser_vacuum": "turbine/condenser_vacuum",
    "circ_water_outlet_temp": "turbine/circ_water_outlet_temp",
    # Meta
    "status": "boiler/status",
    "fault": "system/faults",
}

# ——— NORMAL OPERATING RANGES ———
# Typical values for a subcritical 300-600 MW utility boiler.
NORMAL = {
    "main_steam_flow": {"min": 800, "max": 1000, "mean": 900,
    "unit": "t/h"},
    "main_steam_temp_boiler": {"min": 535, "max": 545, "mean": 540,
    "unit": "C"},
    "main_steam_pressure_boiler": {"min": 16.0, "max": 17.5, "mean": 16.7,
    "unit": "MPa"},
    "reheat_steam_temp_boiler": {"min": 535, "max": 545, "mean": 540,
    "unit": "C"},
    "superheater_desup_flow": {"min": 10, "max": 40, "mean": 25,
    "unit": "t/h"},
    "reheater_desup_flow": {"min": 0, "max": 15, "mean": 5,
    "unit": "t/h"},
    "feedwater_temp": {"min": 270, "max": 285, "mean": 278,
    "unit": "C"},
    "feedwater_flow": {"min": 800, "max": 1000, "mean": 900,
    "unit": "t/h"},
    "feedwater_pressure": {"min": 18.0, "max": 20.0, "mean": 19.0,
    "unit": "MPa"},
    "flue_gas_temp": {"min": 120, "max": 140, "mean": 130,
    "unit": "C"},
    "oxygen_level": {"min": 3, "max": 5, "mean": 4,
    "unit": "%"},
    "main_steam_temp_turbine": {"min": 530, "max": 540, "mean": 535,
    "unit": "C"},
    "main_steam_pressure_turbine": {"min": 15.5, "max": 17.0, "mean": 16.2,
    "unit": "MPa"},
    "reheat_steam_temp_turbine": {"min": 530, "max": 540, "mean": 535,
    "unit": "C"},
    "reheat_steam_pressure_turbine": {"min": 3.0, "max": 4.0, "mean": 3.5,
    "unit": "MPa"},
    "control_stage_pressure": {"min": 10.0, "max": 13.0, "mean": 11.5,
    "unit": "MPa"},
    "high_exhaust_pressure": {"min": 3.0, "max": 4.0, "mean": 3.5,

```

```

"unit": "MPa"},
    "condenser_vacuum":          {"min": 4.0, "max": 7.0, "mean": 5.0,
"unit": "kPa"},
    "circ_water_outlet_temp":    {"min": 25, "max": 35, "mean": 30,
"unit": "C"},
}

# — FAULT DEFINITIONS —
# Each fault picks one sensor and multiplies its mean to push it
# outside the normal range. Multipliers chosen to clearly exceed
# the operating limits in NORMAL above.
FAULT_TYPES = {
    "HIGH_MAIN_STEAM_PRESSURE": {"sensor": "main_steam_pressure_boiler",
"multiplier": 1.15, "severity": "CRITICAL"},
    "LOW_FEEDWATER_FLOW":        {"sensor": "feedwater_flow",
"multiplier": 0.5, "severity": "CRITICAL"},
    "LOW_FEEDWATER_PRESSURE":    {"sensor": "feedwater_pressure",
"multiplier": 0.6, "severity": "CRITICAL"},
    "HIGH_FLUE_GAS_TEMP":        {"sensor": "flue_gas_temp",
"multiplier": 1.35, "severity": "WARNING"},
    "LOW_OXYGEN":                {"sensor": "oxygen_level",
"multiplier": 0.3, "severity": "WARNING"},
    "HIGH_OXYGEN":               {"sensor": "oxygen_level",
"multiplier": 2.0, "severity": "WARNING"},
    "HIGH_REHEAT_TEMP":          {"sensor": "reheat_steam_temp_boiler",
"multiplier": 1.08, "severity": "WARNING"},
    "CONDENSER_VACUUM_LOSS":     {"sensor": "condenser_vacuum",
"multiplier": 3.0, "severity": "CRITICAL"},
    "HIGH_CIRC_WATER_TEMP":      {"sensor": "circ_water_outlet_temp",
"multiplier": 1.3, "severity": "WARNING"},
    "EXCESSIVE_DESUP_SPRAY":     {"sensor": "superheater_desup_flow",
"multiplier": 2.5, "severity": "WARNING"},
}

class BoilerSimulator:
    def __init__(self):
        # Start every sensor at its mean value
        self.state = {name: cfg["mean"] for name, cfg in NORMAL.items()}
        self.active_fault = None
        self.fault_duration = 0
        self.tick = 0 # counts simulation steps

    def _add_noise(self, value, noise_pct=0.01):
        """Add small random noise to simulate sensor jitter."""
        noise = value * noise_pct * random.gauss(0, 1)
        return value + noise

    def _drift_value(self, sensor_name):
        """Make values drift slowly and realistically using sine wave."""
        cfg = NORMAL[sensor_name]
        # Slow oscillation around mean (5% of the sensor's range)
        drift = math.sin(self.tick * 0.05) * (cfg["max"] - cfg["min"]) * 0.05
        return self._add_noise(cfg["mean"] + drift)

    def _inject_fault(self):
        """Randomly decide to start a fault (5% chance per tick)."""
        if self.active_fault is None and random.random() < 0.05:
            fault_name = random.choice(list(FAULT_TYPES.keys()))
            self.active_fault = fault_name
            self.fault_duration = random.randint(10, 30) # lasts 10-30 ticks
            return fault_name
        return None

```

```

def update(self):
    """Update all sensor values for this tick."""
    self.tick += 1

    # Normal drift for every sensor in NORMAL
    for sensor in NORMAL:
        self.state[sensor] = round(self._drift_value(sensor), 3)

    # — Fault logic —————
    new_fault = self._inject_fault()

    if self.active_fault:
        fault = FAULT_TYPES[self.active_fault]
        sensor = fault["sensor"]
        normal_val = NORMAL[sensor]["mean"]
        self.state[sensor] = round(normal_val * fault["multiplier"], 3)

        self.fault_duration -= 1
        if self.fault_duration <= 0:
            self.active_fault = None # Fault resolves

    return new_fault # Return fault name if new fault just started

def get_status(self):
    """Return overall health status."""
    if self.active_fault:
        fault = FAULT_TYPES[self.active_fault]
        return fault["severity"]
    return "NORMAL"

def on_connect(client, userdata, flags, rc):
    if rc == 0:
        print("▣ Boiler simulator connected to MQTT broker")
    else:
        print(f"▣ Connection failed: {rc}")

# ——— MAIN SIMULATION LOOP —————

def main():
    simulator = BoilerSimulator()

    client = mqtt.Client(client_id=CLIENT_ID)
    client.on_connect = on_connect
    client.connect(BROKER_HOST, BROKER_PORT, keepalive=60)
    client.loop_start()

    time.sleep(1)
    print("▣ Boiler simulator started. Publishing every 500ms...")

    while True:
        # Update simulation state
        new_fault = simulator.update()

        # Timestamp for all messages this tick
        timestamp = datetime.utcnow().isoformat() + "Z"

        # Publish each sensor to its own topic
        for sensor_name, topic in TOPICS.items():
            if sensor_name in ("status", "fault"):
                continue # Handle separately

            value = simulator.state.get(sensor_name)

```

```

        if value is None:
            continue

        payload = {
            "device_id": "BOILER_001",
            "sensor": sensor_name,
            "value": value,
            "unit": NORMAL.get(sensor_name, {}).get("unit", ""),
            "timestamp": timestamp,
            "status": simulator.get_status(),
        }
        client.publish(topic, json.dumps(payload), qos=1)

    # Publish status summary
    status_payload = {
        "device_id": "BOILER_001",
        "timestamp": timestamp,
        "overall_status": simulator.get_status(),
        "active_fault": simulator.active_fault,
        "readings": {k: simulator.state[k] for k in NORMAL},
    }
    client.publish(TOPICS["status"], json.dumps(status_payload), qos=1)

    # Publish fault event if new fault just appeared
    if new_fault:
        fault_payload = {
            "device_id": "BOILER_001",
            "fault_code": new_fault,
            "severity": FAULT_TYPES[new_fault]["severity"],
            "affected_sensor": FAULT_TYPES[new_fault]["sensor"],
            "timestamp": timestamp,
            "message": f"Fault detected: {new_fault} on boiler BOILER_001",
        }
        client.publish(TOPICS["fault"], json.dumps(fault_payload), qos=2)
        print(f"△ FAULT: {new_fault} – severity: {FAULT_TYPES[new_fault]['severity']}")

    time.sleep(0.5) # 500ms = 2 messages per second per sensor

if __name__ == "__main__":
    main()

```

Step 2.2 — Chimney Simulator

Create `simulators/chimney_simulator.py`:

```

"""
Chimney Simulator – generates flue gas and draft sensor data.
"""

import paho.mqtt.client as mqtt
import json, time, random, math
from datetime import datetime

BROKER_HOST = "localhost"
BROKER_PORT = 1883
CLIENT_ID = "chimney_simulator"

TOPICS = {

```

```

    "flue_temp":    "chimney/flue_temperature",
    "co2":         "chimney/co2_percentage",
    "o2":         "chimney/o2_percentage",
    "co":         "chimney/co_ppm",
    "draft":       "chimney/draft_pressure",
    "stack_velocity": "chimney/stack_velocity",
    "status":      "chimney/status",
    "fault":       "system/faults",
}

NORMAL = {
    "flue_temp":    {"min": 150, "max": 250, "mean": 200, "unit": "C"},
    "co2":         {"min": 8,   "max": 14,  "mean": 11,  "unit": "%"},
    "o2":         {"min": 3,   "max": 8,   "mean": 5,   "unit": "%"},
    "co":         {"min": 0,   "max": 50,  "mean": 20,  "unit": "ppm"},
    "draft":       {"min": -5,  "max": -2,  "mean": -3.5, "unit": "Pa"},
    "stack_velocity": {"min": 3,   "max": 8,   "mean": 5,   "unit": "m/s"},
}

CHIMNEY_FAULTS = {
    "BLOCKED_FLUE": {"sensor": "draft",    "effect": "reduce", "factor": 0.3,
"severity": "CRITICAL"},
    "HIGH_CO":      {"sensor": "co",        "effect": "spike",  "factor": 5.0,
"severity": "CRITICAL"},
    "LOW_DRAFT":    {"sensor": "draft",    "effect": "reduce", "factor": 0.5,
"severity": "WARNING"},
    "HIGH_FLUE_TEMP": {"sensor": "flue_temp", "effect": "spike",  "factor": 1.4,
"severity": "WARNING"},
}

class ChimneySimulator:
    def __init__(self):
        self.tick = 0
        self.active_fault = None
        self.fault_duration = 0
        self.state = {k: NORMAL[k]["mean"] for k in NORMAL}

    def update(self):
        self.tick += 1
        for sensor in NORMAL:
            cfg = NORMAL[sensor]
            drift = math.sin(self.tick * 0.04 + 1.5) * (cfg["max"] - cfg["min"])
            * 0.05
            noise = random.gauss(0, 1) * cfg["mean"] * 0.015
            self.state[sensor] = round(cfg["mean"] + drift + noise, 2)

        new_fault = None
        if self.active_fault is None and random.random() < 0.04:
            new_fault = random.choice(list(CHIMNEY_FAULTS.keys()))
            self.active_fault = new_fault
            self.fault_duration = random.randint(8, 20)

        if self.active_fault:
            fault = CHIMNEY_FAULTS[self.active_fault]
            sensor = fault["sensor"]
            base = NORMAL[sensor]["mean"]
            if fault["effect"] == "spike":
                self.state[sensor] = round(base * fault["factor"], 2)
            elif fault["effect"] == "reduce":
                self.state[sensor] = round(base * fault["factor"], 2)
            self.fault_duration -= 1
            if self.fault_duration <= 0:
                self.active_fault = None

```

```

        return new_fault

def get_status(self):
    if self.active_fault:
        return CHIMNEY_FAULTS[self.active_fault]["severity"]
    return "NORMAL"

def main():
    sim = ChimneySimulator()
    client = mqtt.Client(client_id=CLIENT_ID)
    client.connect(BROKER_HOST, BROKER_PORT, keepalive=60)
    client.loop_start()
    time.sleep(1)
    print("🏠 Chimney simulator started...")

    while True:
        new_fault = sim.update()
        ts = datetime.utcnow().isoformat() + "Z"

        for sensor_name, topic in TOPICS.items():
            if sensor_name in ("status", "fault"):
                continue
            payload = {
                "device_id": "CHIMNEY_001",
                "sensor": sensor_name,
                "value": sim.state.get(sensor_name),
                "unit": NORMAL.get(sensor_name, {}).get("unit", ""),
                "timestamp": ts,
                "status": sim.get_status(),
            }
            client.publish(topic, json.dumps(payload), qos=1)

        if new_fault:
            fault_payload = {
                "device_id": "CHIMNEY_001",
                "fault_code": new_fault,
                "severity": CHIMNEY_FAULTS[new_fault]["severity"],
                "timestamp": ts,
                "message": f"Chimney fault: {new_fault}",
            }
            client.publish(TOPICS["fault"], json.dumps(fault_payload), qos=2)
            print(f"⚠️ CHIMNEY FAULT: {new_fault}")

        time.sleep(0.5)

if __name__ == "__main__":
    main()

```

☐ Day 2 Checklist

- ☐ `boiler_simulator.py` runs and publishes to 8 topics
- ☐ `chimney_simulator.py` runs and publishes to 7 topics
- ☐ Fault messages appear occasionally on `system/faults`
- ☐ You can see messages in EMQX dashboard under Topics tab

- ☐ Both simulators run simultaneously in separate terminals
-

DAY 3 — InfluxDB + Grafana (Live Dashboard)

☐ Goal

By end of Day 3:

- All MQTT sensor data automatically saved to InfluxDB
 - A Grafana dashboard shows live charts of boiler and chimney sensors
 - You can query historical data and see fault events on the timeline
-

Concept: Why InfluxDB Instead of Regular SQL?

A regular SQL database stores rows like:

id	timestamp	sensor_name	value
1	2024-01-01 10:00:00	temperature	85.4
2	2024-01-01 10:00:01	temperature	85.6

InfluxDB stores **measurements** optimised for time queries:

```
measurement: boiler_sensors
tags: device_id=BOILER_001, sensor=temperature
fields: value=85.4
time: 2024-01-01T10:00:00Z
```

Why it matters: querying "average temperature over the last 5 minutes" on InfluxDB takes milliseconds on millions of rows. On SQL, it can take seconds.

Step 3.1 — Add InfluxDB and Grafana to Docker Compose

Update your `docker-compose.yml`:

```
version: '3.8'

services:
  emqx:
    image: emqx/emqx:latest
    container_name: emqx_broker
    ports:
      - "1883:1883"
      - "8083:8083"
      - "18083:18083"
    environment:
      - EMQX_ALLOW_ANONYMOUS=true
    restart: unless-stopped

  influxdb:
```

```

image: influxdb:2.7
container_name: influxdb
ports:
  - "8086:8086"
environment:
  - DOCKER_INFLUXDB_INIT_MODE=setup
  - DOCKER_INFLUXDB_INIT_USERNAME=admin
  - DOCKER_INFLUXDB_INIT_PASSWORD=password123
  - DOCKER_INFLUXDB_INIT_ORG=boiler_org
  - DOCKER_INFLUXDB_INIT_BUCKET=boiler_data
  - DOCKER_INFLUXDB_INIT_ADMIN_TOKEN=my-super-secret-token-123
volumes:
  - influxdb_data:/var/lib/influxdb2
restart: unless-stopped

```

```

grafana:
  image: grafana/grafana:latest
  container_name: grafana
  ports:
    - "3000:3000"
  environment:
    - GF_SECURITY_ADMIN_USER=admin
    - GF_SECURITY_ADMIN_PASSWORD=admin123
  volumes:
    - grafana_data:/var/lib/grafana
  depends_on:
    - influxdb
  restart: unless-stopped

```

```

volumes:
  influxdb_data:
  grafana_data:

```

```
docker-compose up -d
```

Step 3.2 — MQTT to InfluxDB Bridge (Consumer)

Install dependencies:

```
pip install influxdb-client paho-mqtt
```

Create consumers/influx_consumer.py:

```

"""
InfluxDB Consumer – subscribes to ALL MQTT topics and writes every
sensor reading into InfluxDB as a time-series data point.

This is Consumer #1 in your producer/consumer architecture.
"""

import paho.mqtt.client as mqtt
import json
from datetime import datetime
from influxdb_client import InfluxDBClient, Point, WritePrecision
from influxdb_client.client.write_api import SYNCHRONOUS

# — CONFIG —
MQTT_HOST = "localhost"
MQTT_PORT = 1883

```



```

INFLUX_URL    = "http://localhost:8086"
INFLUX_TOKEN  = "my-super-secret-token-123"
INFLUX_ORG    = "boiler_org"
INFLUX_BUCKET = "boiler_data"
# -----

# Connect to InfluxDB
influx_client = InfluxDBClient(url=INFLUX_URL, token=INFLUX_TOKEN,
                                org=INFLUX_ORG)
write_api = influx_client.write_api(write_options=SYNCHRONOUS)

print("▯ Connected to InfluxDB")

def on_connect(client, userdata, flags, rc):
    if rc == 0:
        print("▯ InfluxDB consumer connected to MQTT broker")
        # Subscribe to ALL boiler and chimney topics
        client.subscribe("boiler/#", qos=1)
        client.subscribe("chimney/#", qos=1)
        client.subscribe("system/faults", qos=2)
        print("👉 Subscribed to boiler/#, chimney/#, system/faults")

def on_message(client, userdata, message):
    """Every MQTT message → write a data point to InfluxDB."""
    try:
        payload = json.loads(message.payload.decode("utf-8"))
        topic = message.topic

        # Determine measurement name from topic prefix
        if topic.startswith("boiler"):
            measurement = "boiler_sensors"
        elif topic.startswith("chimney"):
            measurement = "chimney_sensors"
        elif topic == "system/faults":
            measurement = "fault_events"
        else:
            return

        device_id = payload.get("device_id", "UNKNOWN")

        # — Write sensor reading —————
        if "value" in payload:
            point = (
                Point(measurement)
                .tag("device_id", device_id)
                .tag("sensor", payload.get("sensor", topic.split("/")[-1]))
                .tag("status", payload.get("status", "NORMAL"))
                .field("value", float(payload["value"]))
                .time(payload.get("timestamp", datetime.utcnow().isoformat()))
            )
            write_api.write(bucket=INFLUX_BUCKET, org=INFLUX_ORG, record=point)
            print(f"▯ Stored: {device_id}/{payload.get('sensor')} = {payload['value']}")

        # — Write fault event —————
        elif "fault_code" in payload:
            point = (
                Point("fault_events")
                .tag("device_id", device_id)
                .tag("fault_code", payload["fault_code"])
                .tag("severity", payload.get("severity", "UNKNOWN"))
                .field("message", payload.get("message", ""))
            )

```

```

        .field("affected_sensor", payload.get("affected_sensor", ""))
        .time(payload.get("timestamp", datetime.utcnow().isoformat()))
    )
    write_api.write(bucket=INFLUX_BUCKET, org=INFLUX_ORG, record=point)
    print(f"    Fault stored: {payload['fault_code']}")
    ({payload.get('severity')})")

except Exception as e:
    print(f"    Error processing message: {e}")

# Setup MQTT client
mqtt_client = mqtt.Client(client_id="influx_consumer")
mqtt_client.on_connect = on_connect
mqtt_client.on_message = on_message
mqtt_client.connect(MQTT_HOST, MQTT_PORT, keepalive=60)

print("🔊 InfluxDB consumer listening...")
mqtt_client.loop_forever()

```

Step 3.3 — Set Up Grafana Dashboard

1. Open <http://localhost:3000> — login: admin / admin123
2. Go to **Configuration** → **Data Sources** → **Add data source**
3. Choose **InfluxDB**
4. Set:
 - Query Language: **Flux**
 - URL: <http://influxdb:8086>
 - Organization: **boiler_org**
 - Token: **my-super-secret-token-123**
 - Default Bucket: **boiler_data**
5. Click **Save & Test** → should say "datasource is working"

Add a panel — Boiler Temperature:

Click + → **New Dashboard** → **Add Panel**

Paste this Flux query:

```

from(bucket: "boiler_data")
  |> range(start: -10m)
  |> filter(fn: (r) => r["_measurement"] == "boiler_sensors")
  |> filter(fn: (r) => r["sensor"] == "temperature")
  |> filter(fn: (r) => r["_field"] == "value")

```

Set refresh to **5s** and you will see live temperature data.

☐ Day 3 Checklist

- ☐ InfluxDB running, accessible at localhost:8086
 - ☐ Grafana running, accessible at localhost:3000
 - ☐ `influx_consumer.py` running and storing data
 - ☐ Grafana dashboard shows live temperature chart
 - ☐ Fault events visible in InfluxDB
-
-

DAY 4 — Kafka Buffer + JSONL Training Data Exporter

☐ Goal

By end of Day 4:

- Kafka handles 25,000+ messages/second as a buffer layer
 - A JSONL exporter generates fine-tuning training data from stored sensor readings
 - You have a proper `train.jsonl` file ready for LLM fine-tuning
-

Concept: Why Kafka for 25K msg/s?

Without Kafka:

Simulators → MQTT Broker → InfluxDB consumer (can write ~1000 rows/sec)

At 25K messages/second, InfluxDB consumer falls behind. Messages pile up and eventually drop.

With Kafka:

Simulators → MQTT Broker → Kafka Topic (buffer, handles millions/sec)
↓ (consumers read at their own pace)

InfluxDB	Fault Det.	Vector DB	JSONL
(1K/sec)	(5K/sec)	(2K/sec)	(500/sec)

Kafka is a **durable log** — messages are stored on disk for 7 days by default. Even if your InfluxDB goes down, you don't lose data — it catches up when it comes back.

Step 4.1 — Add Kafka to Docker Compose

Add to your `docker-compose.yml`:

```
zookeeper:
  image: confluentinc/cp-zookeeper:7.5.0
  container_name: zookeeper
```

```

environment:
  ZOOKEEPER_CLIENT_PORT: 2181
restart: unless-stopped

kafka:
  image: confluentinc/cp-kafka:7.5.0
  container_name: kafka
  ports:
    - "9092:9092"
  environment:
    KAFKA_BROKER_ID: 1
    KAFKA_ZOOKEEPER_CONNECT: zookeeper:2181
    KAFKA_ADVERTISED_LISTENERS: PLAINTEXT://localhost:9092
    KAFKA_OFFSETS_TOPIC_REPLICATION_FACTOR: 1
    KAFKA_AUTO_CREATE_TOPICS_ENABLE: "true"
  depends_on:
    - zookeeper
  restart: unless-stopped

```

```

docker-compose up -d kafka zookeeper
pip install kafka-python

```

Step 4.2 — MQTT → Kafka Bridge

Create consumers/kafka_bridge.py:

```

"""
Kafka Bridge – consumes all MQTT messages and forwards to Kafka topics.
This is the entry point for the high-throughput pipeline.
"""

import paho.mqtt.client as mqtt
from kafka import KafkaProducer
import json, time

MQTT_HOST = "localhost"
KAFKA_BOOTSTRAP = "localhost:9092"

# Kafka topics (one per device/type for parallel processing)
KAFKA_TOPICS = {
    "boiler": "boiler-sensors",
    "chimney": "chimney-sensors",
    "faults": "system-faults",
}

producer = KafkaProducer(
    bootstrap_servers=KAFKA_BOOTSTRAP,
    value_serializer=lambda v: json.dumps(v).encode("utf-8"),
    acks=1, # wait for leader acknowledgment
    compression_type="gzip", # compress for throughput
)

def on_connect(client, userdata, flags, rc):
    if rc == 0:
        client.subscribe("boiler/#", qos=1)
        client.subscribe("chimney/#", qos=1)
        client.subscribe("system/faults", qos=2)
        print("Kafka bridge connected, subscribed to all topics")

def on_message(client, userdata, msg):

```

```

try:
    payload = json.loads(msg.payload.decode("utf-8"))

    # Route to correct Kafka topic
    if msg.topic.startswith("boiler"):
        kafka_topic = KAFKA_TOPICS["boiler"]
    elif msg.topic.startswith("chimney"):
        kafka_topic = KAFKA_TOPICS["chimney"]
    else:
        kafka_topic = KAFKA_TOPICS["faults"]

    # Send to Kafka (non-blocking, batched internally)
    producer.send(kafka_topic, value=payload)

except Exception as e:
    print(f"❏ Kafka bridge error: {e}")

client = mqtt.Client(client_id="kafka_bridge")
client.on_connect = on_connect
client.on_message = on_message
client.connect(MQTT_HOST, 1883)
print("❏ Kafka bridge running...")
client.loop_forever()

```

Step 4.3 — JSONL Training Data Exporter

This is critical — it converts your InfluxDB data into the Q&A pairs your LLM needs.

Create `exporters/jsonl_exporter.py`:

```

"""
JSONL Exporter – reads recent sensor data from InfluxDB and generates
question-answer pairs for LLM fine-tuning.

Output format: one JSON object per line, Alpaca/ChatML style.
"""

import json
from datetime import datetime, timedelta
from influxdb_client import InfluxDBClient

INFLUX_URL = "http://localhost:8086"
INFLUX_TOKEN = "my-super-secret-token-123"
INFLUX_ORG = "boiler_org"
INFLUX_BUCKET = "boiler_data"
OUTPUT_FILE = "data/train.jsonl"

client = InfluxDBClient(url=INFLUX_URL, token=INFLUX_TOKEN, org=INFLUX_ORG)
query_api = client.query_api()

def query_recent_readings(sensor: str, device: str = "BOILER_001", minutes: int = 60):
    """Get last N minutes of readings for a sensor."""
    query = f"""
    from(bucket: "{INFLUX_BUCKET}")
      |> range(start: -{minutes}m)
      |> filter(fn: (r) => r["_measurement"] == "boiler_sensors")
      |> filter(fn: (r) => r["sensor"] == "{sensor}")
      |> filter(fn: (r) => r["device_id"] == "{device}")
    """

```

```

    |> filter(fn: (r) => r["_field"] == "value")
    |> mean()
    """
    tables = query_api.query(query)
    for table in tables:
        for record in table.records:
            return round(record.get_value(), 2)
    return None

def query_fault_history(hours: int = 24):
    """Get fault events from the last N hours."""
    query = f"""
    from(bucket: "{INFLUX_BUCKET}")
    |> range(start: -{hours}h)
    |> filter(fn: (r) => r["_measurement"] == "fault_events")
    |> filter(fn: (r) => r["_field"] == "message")
    """
    faults = []
    tables = query_api.query(query)
    for table in tables:
        for record in table.records:
            faults.append({
                "time": str(record.get_time()),
                "fault_code": record.values.get("fault_code"),
                "severity": record.values.get("severity"),
                "message": record.get_value(),
            })
    return faults

def generate_qa_pairs():
    """Generate diverse Q&A pairs from live data."""
    pairs = []

    # — Sensor readings Q&A —————
    sensors = ["temperature", "pressure", "fuel_flow", "water_level"]
    for sensor in sensors:
        value = query_recent_readings(sensor)
        if value is None:
            continue

        # Normal operating ranges
        ranges = {
            "temperature": (60, 100, "°C"),
            "pressure": (10, 14, "bar"),
            "fuel_flow": (5, 20, "L/min"),
            "water_level": (40, 60, "%"),
        }

        lo, hi, unit = ranges[sensor]
        status = "within normal range" if lo <= value <= hi else "OUTSIDE NORMAL
RANGE – requires attention"

        pairs.append({
            "messages": [
                {
                    "role": "user", "content": f"What is the current boiler
{sensor.replace('_', ' ')}?"},
                {
                    "role": "assistant", "content": f"The current boiler
{sensor.replace('_', ' ')} is {value} {unit}, which is {status} (normal: {lo}-
{hi} {unit})."}
            ]
        })

```

```

        pairs.append({
            "messages": [
                {"role": "user", "content": f"Is the boiler {sensor.replace('_',
' ')} normal?"},
                {"role": "assistant", "content": f"The boiler
{sensor.replace('_', ' ')} is currently {value} {unit}. The safe operating range
is {lo}-{hi} {unit}. Current reading is {status}."}
            ]
        })

# — Fault history Q&A —————
faults = query_fault_history(hours=24)
if faults:
    fault_summary = "; ".join([f"{f['fault_code']} at {f['time']}" for f in
faults[:5]])
    pairs.append({
        "messages": [
            {"role": "user", "content": "What faults have occurred in the
last 24 hours?"},
            {"role": "assistant", "content": f"In the last 24 hours, the
following faults were recorded: {fault_summary}. Total fault count:
{len(faults)}."}
        ]
    })

    critical = [f for f in faults if f["severity"] == "CRITICAL"]
    if critical:
        pairs.append({
            "messages": [
                {"role": "user", "content": "Are there any critical
faults?"},
                {"role": "assistant", "content": f"Yes, {len(critical)}
critical fault(s) detected: " + "; ".join([f['fault_code'] for f in critical]) +
". Immediate attention required."}
            ]
        })
    else:
        pairs.append({
            "messages": [
                {"role": "user", "content": "Are there any active faults?"},
                {"role": "assistant", "content": "No faults detected in the last
24 hours. All boiler and chimney systems are operating normally."}
            ]
        })

    return pairs

# — Main export —————
import os
os.makedirs("data", exist_ok=True)

print("\n Generating JSONL training data from InfluxDB...")
qa_pairs = generate_qa_pairs()

with open(OUTPUT_FILE, "w") as f:
    for pair in qa_pairs:
        f.write(json.dumps(pair) + "\n")

print(f" Exported {len(qa_pairs)} Q&A pairs to {OUTPUT_FILE}")

```

☐ Day 4 Checklist

- ☐ Kafka running in Docker
 - ☐ `kafka_bridge.py` forwarding MQTT → Kafka
 - ☐ `jsonl_exporter.py` generates `data/train.jsonl` with real Q&A pairs
 - ☐ File has at least 50+ pairs (run for a few hours first)
-
-

DAY 5 — Fine-Tuning the LLM with LoRA

☐ Goal

Fine-tune Mistral 7B or Llama 3.1 8B on your boiler training data using LoRA, so the model understands your domain.

Concept: Fine-Tuning vs RAG — When to Use Each

	Fine-Tuning	RAG
What it does	Bakes domain knowledge into model weights	Fetches relevant context at query time
Good for	Domain vocabulary, fault code definitions, equipment-specific reasoning	Real-time data, latest readings, recent faults
Updates	Requires retraining	Immediate — just update the vector DB
You need both	Fine-tune for <i>how to reason</i> about boiler data	RAG for <i>what the current data says</i>

You are building a **hybrid system**: fine-tuning teaches the model what CO₂ at 18% means for a chimney; RAG tells it that CO₂ right now is 18%.

Concept: LoRA — Low Rank Adaptation

A 7B parameter model has 7 billion numbers in its weights. Full fine-tuning changes all of them — requires 8+ GPUs.

LoRA adds two small matrices (A and B) next to each weight matrix. Only A and B are trained. This is ~0.1% of the total parameters.

Original weight W (frozen)
+
LoRA adapter: $A \times B$ (trained)

=
Effective weight: $W + A \times B$

You can fine-tune a 7B model on a single RTX 3090 / A100 GPU in 2-4 hours.

Step 5.1 — Install Fine-Tuning Dependencies

```
pip install transformers peft datasets accelerate bitsandbytes torch
```

Create training/fine_tune.py:

```
"""
Fine-tuning script using LoRA (QLoRA for 4-bit quantization).
Trains Mistral 7B on your boiler Q&A dataset.

Requirements:
- GPU with 16GB+ VRAM (RTX 3090 / A100 / T4)
- Or use Google Colab A100 (free tier available)
"""

import json
import torch
from datasets import Dataset
from transformers import (
    AutoModelForCausalLM,
    AutoTokenizer,
    TrainingArguments,
    Trainer,
    BitsAndBytesConfig,
)
from peft import get_peft_model, LoraConfig, TaskType,
prepare_model_for_kbit_training

# — CONFIG —
BASE_MODEL = "mistralai/Mistral-7B-Instruct-v0.2" # Or "meta-llama/Llama-3.1-8B-Instruct"
TRAIN_FILE = "../data/train.jsonl"
OUTPUT_DIR = "../models/boiler_llm"
MAX_LENGTH = 512

# LoRA hyperparameters
LORA_R = 16 # Rank — higher = more capacity but more memory
LORA_ALPHA = 32 # Scaling factor (usually 2x rank)
LORA_DROPOUT = 0.05

#

def load_jsonl(filepath):
    """Load JSONL file into a list of dicts."""
    records = []
    with open(filepath) as f:
        for line in f:
            records.append(json.loads(line.strip()))
    return records

def format_prompt(example):
    """Convert ChatML format to model's expected format."""
    messages = example["messages"]
    user_msg = next(m["content"] for m in messages if m["role"] == "user")
```

```

    asst_msg = next(m["content"] for m in messages if m["role"] == "assistant")

    # Mistral instruction format
    text = f"<s>[INST] {user_msg} [/INST] {asst_msg} </s>"
    return {"text": text}

# — 1. Load and prepare dataset —————
print("▢ Loading training data...")
raw_data = load_jsonl(TRAIN_FILE)
dataset = Dataset.from_list(raw_data).map(format_prompt)
dataset = dataset.train_test_split(test_size=0.1)
print(f"▢ Dataset: {len(dataset['train'])} train, {len(dataset['test'])} eval samples")

# — 2. Load tokenizer —————
print("▢ Loading tokenizer...")
tokenizer = AutoTokenizer.from_pretrained(BASE_MODEL, trust_remote_code=True)
tokenizer.pad_token = tokenizer.eos_token
tokenizer.padding_side = "right"

# — 3. Load model in 4-bit (QLoRA) —————
print("▢ Loading model in 4-bit quantization (QLoRA)...")
bnb_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_use_double_quant=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.bfloat16,
)

model = AutoModelForCausalLM.from_pretrained(
    BASE_MODEL,
    quantization_config=bnb_config,
    device_map="auto",
    trust_remote_code=True,
)
model = prepare_model_for_kbit_training(model)

# — 4. Configure LoRA —————
print("▢ Applying LoRA adapters...")
lora_config = LoraConfig(
    task_type=TaskType.CAUSAL_LM,
    r=LORA_R,
    lora_alpha=LORA_ALPHA,
    lora_dropout=LORA_DROPOUT,
    target_modules=["q_proj", "v_proj", "k_proj", "o_proj"], # Attention
    matrices
    bias="none",
)
model = get_peft_model(model, lora_config)
model.print_trainable_parameters()
# Output: trainable params: 4,194,304 || all params: 3,756,040,192 ||
trainable%: 0.11%

# — 5. Tokenize —————
def tokenize(example):
    return tokenizer(
        example["text"],
        truncation=True,
        max_length=MAX_LENGTH,
        padding="max_length",
    )

tokenized = dataset.map(tokenize, batched=True)

```

```
# — 6. Training arguments —————
training_args = TrainingArguments(
    output_dir=OUTPUT_DIR,
    num_train_epochs=3,
    per_device_train_batch_size=4,
    gradient_accumulation_steps=4,      # Effective batch = 4 × 4 = 16
    learning_rate=2e-4,
    fp16=True,
    save_steps=100,
    logging_steps=25,
    evaluation_strategy="steps",
    eval_steps=100,
    warmup_steps=50,
    report_to="none",
)

# — 7. Train —————
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized["train"],
    eval_dataset=tokenized["test"],
)

print("▣ Starting fine-tuning...")
trainer.train()
model.save_pretrained(OUTPUT_DIR)
tokenizer.save_pretrained(OUTPUT_DIR)
print(f"▣ Fine-tuned model saved to {OUTPUT_DIR}")
```

No GPU? Upload `train.jsonl` to Google Colab and run there. Free T4 is sufficient for this dataset size.

☐ Day 5 Checklist

- ☐ Fine-tuning dependencies installed
 - ☐ `train.jsonl` has 100+ Q&A pairs
 - ☐ Fine-tuning runs without OOM errors
 - ☐ Model saved to `models/boiler_llm/`
-

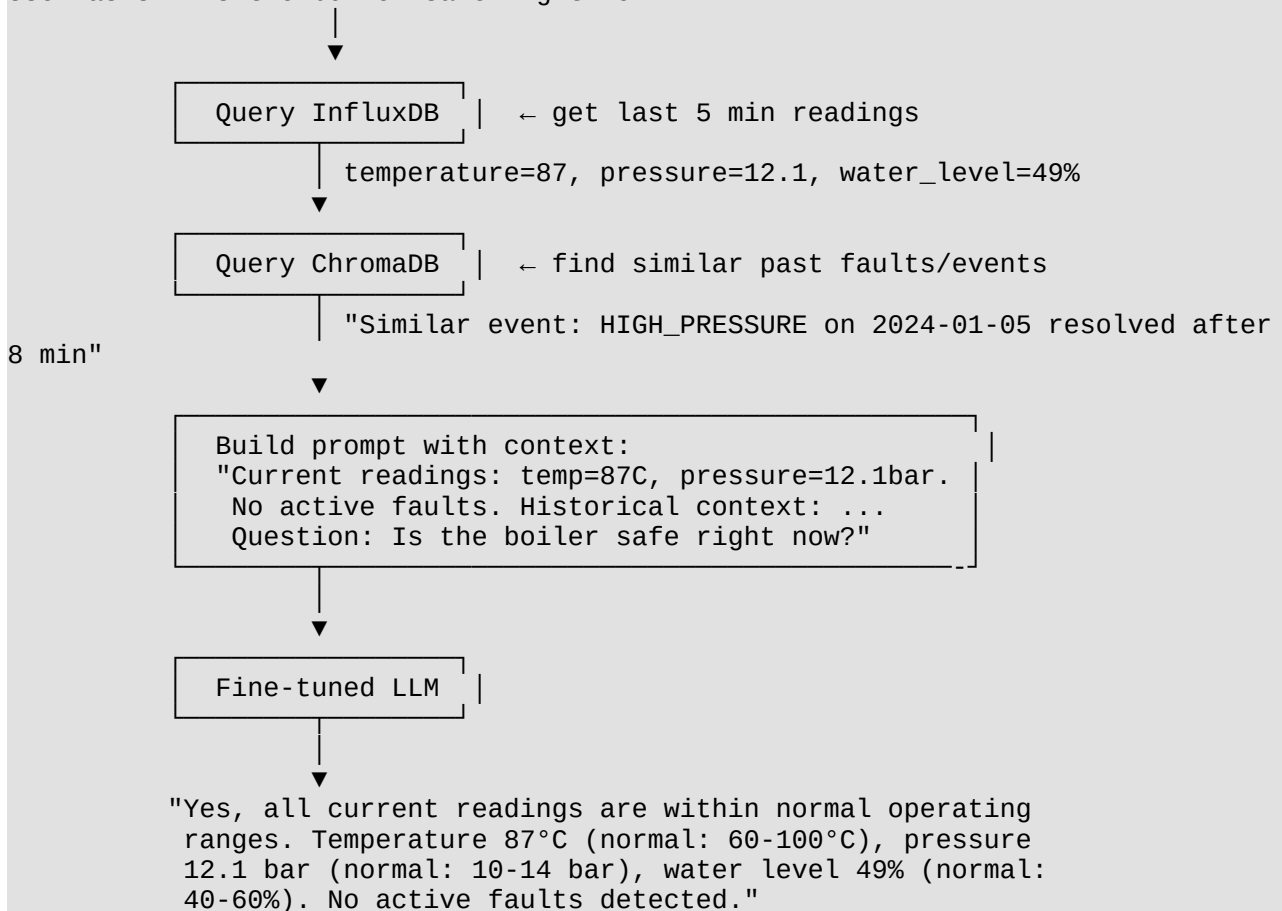
DAY 6 — RAG Pipeline with ChromaDB + LangChain

☐ Goal

Build a RAG pipeline that fetches the latest sensor readings before every LLM query, so the chatbot always answers from real current data.

Concept: How RAG Works in This System

User asks: "Is the boiler safe right now?"



Step 6.1 — Install RAG Dependencies

```
pip install langchain langchain-community chromadb sentence-transformers ollama
```

Serve your fine-tuned model with Ollama:

```
# Install Ollama from https://ollama.ai
# Then create a Modelfile
cat > Modelfile << EOF
FROM ./models/boiler_llm
SYSTEM "You are a boiler and chimney monitoring assistant. You only answer
questions based on real-time sensor data and fault history provided to you. Do
not use general knowledge."
EOF

ollama create boiler-assistant -f Modelfile
ollama serve # starts model server on localhost:11434
```

Step 6.2 — RAG Pipeline

Create rag/rag_pipeline.py:

```
"""
```

RAG Pipeline – connects InfluxDB (real-time data), ChromaDB (semantic search), and the fine-tuned LLM to answer boiler/chimney questions grounded in real data.
"""

```
from langchain.llms import Ollama
from langchain.embeddings import HuggingFaceEmbeddings
from langchain.vectorstores import Chroma
from langchain.prompts import PromptTemplate
from langchain.chains import RetrievalQA
from influxdb_client import InfluxDBClient
import chromadb
import json
from datetime import datetime
```

```
# — CONFIG —
INFLUX_URL = "http://localhost:8086"
INFLUX_TOKEN = "my-super-secret-token-123"
INFLUX_ORG = "boiler_org"
INFLUX_BUCKET = "boiler_data"
CHROMA_PATH = "./chroma_db"
OLLAMA_MODEL = "boiler-assistant"
#
```

```
class BoilerRAGPipeline:
    def __init__(self):
        print("▣ Initializing RAG pipeline...")

        # InfluxDB client
        self.influx = InfluxDBClient(
            url=INFLUX_URL, token=INFLUX_TOKEN, org=INFLUX_ORG
        )
        self.query_api = self.influx.query_api()

        # Embedding model (converts text to vectors for semantic search)
        self.embeddings = HuggingFaceEmbeddings(
            model_name="sentence-transformers/all-MiniLM-L6-v2"
        )

        # ChromaDB vector store
        self.vectorstore = Chroma(
            persist_directory=CHROMA_PATH,
            embedding_function=self.embeddings
        )

        # Fine-tuned LLM via Ollama
        self.llm = Ollama(model=OLLAMA_MODEL, temperature=0.1)

        print("▣ RAG pipeline ready")

    def get_realtime_context(self):
        """Fetch latest readings from InfluxDB as context string."""
        context_parts = []

        sensors = {
            "boiler_sensors": ["temperature", "pressure", "fuel_flow",
"water_level"],
            "chimney_sensors": ["flue_temp", "co2", "co", "draft"],
        }

        for measurement, sensor_list in sensors.items():
            device = "BOILER_001" if "boiler" in measurement else "CHIMNEY_001"
            for sensor in sensor_list:
                query = f"""
```

```

        from(bucket: "{INFLUX_BUCKET}")
        |> range(start: -5m)
        |> filter(fn: (r) => r["_measurement"] == "{measurement}")
        |> filter(fn: (r) => r["sensor"] == "{sensor}")
        |> filter(fn: (r) => r["_field"] == "value")
        |> last()
    """
    try:
        tables = self.query_api.query(query)
        for table in tables:
            for record in table.records:
                val = round(record.get_value(), 2)
                status_tag = record.values.get("status", "NORMAL")
                context_parts.append(
                    f"{device}/{sensor}: {val} [{status_tag}]"
                )
    except Exception:
        pass

# Get recent faults
fault_query = """
from(bucket: "{INFLUX_BUCKET}")
    |> range(start: -1h)
    |> filter(fn: (r) => r["_measurement"] == "fault_events")
    |> filter(fn: (r) => r["_field"] == "message")
    |> last()
"""
try:
    tables = self.query_api.query(fault_query)
    for table in tables:
        for record in table.records:
            context_parts.append(
                f"RECENT FAULT: {record.values.get('fault_code')} "
                f"({record.values.get('severity')}) at "
                f"{record.get_time()}"
            )
except Exception:
    pass

return "\n".join(context_parts) if context_parts else "No recent data
available."

def index_fault_history(self):
    """Index fault history into ChromaDB for semantic retrieval."""
    print("🔍 Indexing fault history into ChromaDB...")
    query = """
from(bucket: "{INFLUX_BUCKET}")
    |> range(start: -7d)
    |> filter(fn: (r) => r["_measurement"] == "fault_events")
    |> filter(fn: (r) => r["_field"] == "message")
    """
    docs, metas, ids = [], [], []
    tables = self.query_api.query(query)
    for table in tables:
        for i, record in enumerate(table.records):
            text = (
                f"Fault: {record.values.get('fault_code')} "
                f"Severity: {record.values.get('severity')} "
                f"Time: {record.get_time()} "
                f"Message: {record.get_value()}"
            )
            docs.append(text)
            metas.append({"type": "fault", "time": str(record.get_time())})
            ids.append(f"fault_{i}")

```

```

    if docs:
        self.vectorstore.add_texts(docs, metadatas=metas, ids=ids)
        self.vectorstore.persist()
        print(f"▢ Indexed {len(docs)} fault records into ChromaDB")

def answer(self, question: str) -> str:
    """Answer a question using real-time data + RAG."""

    # 1. Get real-time context from InfluxDB
    realtime_context = self.get_realtime_context()

    # 2. Retrieve semantically similar past events from ChromaDB
    similar_docs = self.vectorstore.similarity_search(question, k=3)
    historical_context = "\n".join([doc.page_content for doc in
similar_docs])

    # 3. Build the full prompt
    prompt = f"""You are a boiler and chimney monitoring assistant.
Answer the question based ONLY on the real-time sensor data and fault history
below.
Do not use any general knowledge. If data is not available, say so.

=== CURRENT REAL-TIME READINGS ===
{realtime_context}

=== RELEVANT HISTORICAL EVENTS ===
{historical_context}

=== QUESTION ===
{question}

=== ANSWER ==="""

    # 4. Get LLM response
    response = self.llm(prompt)
    return response.strip()

if __name__ == "__main__":
    pipeline = BoilerRAGPipeline()

    # Index fault history (do this once, then incrementally)
    pipeline.index_fault_history()

    # Test it
    test_questions = [
        "What is the current boiler pressure?",
        "Are there any active faults?",
        "Is the boiler safe to operate?",
        "What was the last fault on the chimney?",
    ]

    for q in test_questions:
        print(f"\n▢ Q: {q}")
        print(f"▢ A: {pipeline.answer(q)}")

```

▢ Day 6 Checklist

- ☐ ChromaDB set up and fault history indexed

- ☐ Ollama serving the fine-tuned model
 - ☐ RAG pipeline fetches real-time InfluxDB data as context
 - ☐ Test questions get answers grounded in actual sensor data
-
-

DAY 7 — FastAPI Chatbot + Full Integration

☐ Goal

Build the final chatbot API and UI. Wire everything together. End-to-end: boiler → MQTT → InfluxDB → RAG → chatbot response.

Step 7.1 — FastAPI Chatbot Server

```
pip install fastapi uvicorn websockets
```

Create `api/chatbot_api.py`:

```
"""
FastAPI Chatbot API – exposes the RAG pipeline as REST + WebSocket endpoints.
"""

from fastapi import FastAPI, WebSocket
from fastapi.middleware.cors import CORSMiddleware
from fastapi.staticfiles import StaticFiles
from pydantic import BaseModel
import sys
sys.path.append("..")
from rag_pipeline import BoilerRAGPipeline
import json
from datetime import datetime

app = FastAPI(title="Boiler & Chimney AI Chatbot", version="1.0")

app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_methods=["*"],
    allow_headers=["*"],
)

# Initialize RAG pipeline (takes ~30 seconds on first load)
pipeline = BoilerRAGPipeline()

class ChatRequest(BaseModel):
    question: str
    session_id: str = "default"

class ChatResponse(BaseModel):
    answer: str
    timestamp: str
    question: str
```



```

@app.get("/")
async def root():
    return {"status": "Boiler Chatbot API running", "version": "1.0"}

@app.get("/health")
async def health():
    return {"status": "healthy", "timestamp": datetime.utcnow().isoformat()}

@app.post("/chat", response_model=ChatResponse)
async def chat(request: ChatRequest):
    """Main chat endpoint – accepts a question, returns grounded answer."""
    answer = pipeline.answer(request.question)
    return ChatResponse(
        answer=answer,
        timestamp=datetime.utcnow().isoformat(),
        question=request.question,
    )

@app.get("/realtime")
async def realtime_data():
    """Return latest sensor readings as JSON."""
    context = pipeline.get_realtime_context()
    return {"data": context, "timestamp": datetime.utcnow().isoformat()}

@app.websocket("/ws/chat")
async def websocket_chat(websocket: WebSocket):
    """WebSocket endpoint for streaming chat responses."""
    await websocket.accept()
    try:
        while True:
            question = await websocket.receive_text()
            answer = pipeline.answer(question)
            await websocket.send_json({
                "answer": answer,
                "timestamp": datetime.utcnow().isoformat()
            })
    except Exception:
        pass

```

Run it:

```

cd api
uvicorn chatbot_api:app --host 0.0.0.0 --port 8000 --reload

```

API docs auto-generated at: <http://localhost:8000/docs>

Step 7.2 — Run the Complete System

Create `start_system.sh`:

```

#!/bin/bash
echo "▣ Starting Boiler IoT + AI System..."

# 1. Start all Docker services
docker-compose up -d
sleep 10

# 2. Start simulators (background)
python simulators/boiler_simulator.py &
python simulators/chimney_simulator.py &

```

```
# 3. Start consumers (background)
python consumers/influx_consumer.py &
python consumers/kafka_bridge.py &

# 4. Start chatbot API
cd api && uvicorn chatbot_api:app --host 0.0.0.0 --port 8000

echo "▣ All services running!"
echo "Grafana:      http://localhost:3000"
echo "EMQX:        http://localhost:18083"
echo "InfluxDB:     http://localhost:8086"
echo "Chatbot API:  http://localhost:8000"
echo "API Docs:    http://localhost:8000/docs"
```

Step 7.3 — Test the Full System

```
# Test the chatbot via curl
curl -X POST http://localhost:8000/chat \
  -H "Content-Type: application/json" \
  -d '{"question": "What is the current boiler pressure and is it safe?"}'

# Get real-time readings
curl http://localhost:8000/realtime
```

☐ Day 7 Checklist

- ☐ FastAPI server running at localhost:8000
 - ☐ /chat endpoint returns real-data-grounded answers
 - ☐ /realtime endpoint returns live sensor readings
 - ☐ WebSocket endpoint working for streaming chat
 - ☐ All services running simultaneously without errors
-
-

☐ Final Project Folder Structure

```
boiler-iot-system/
├── docker-compose.yml      ← All infrastructure (EMQX, InfluxDB, Grafana, Kafka)
├── simulators/
│   ├── boiler_simulator.py ← Boiler sensor producer
│   └── chimney_simulator.py ← Chimney sensor producer
├── consumers/
│   ├── influx_consumer.py  ← MQTT → InfluxDB writer
│   └── kafka_bridge.py     ← MQTT → Kafka buffer
```

exporters/ └─ jsonl_exporter.py	← InfluxDB → train.jsonl
data/ └─ train.jsonl	← Fine-tuning dataset
training/ └─ fine_tune.py	← LoRA fine-tuning script
models/ └─ boiler_llm/	← Saved fine-tuned model
rag/ └─ rag_pipeline.py └─ chroma_db/	← LangChain + ChromaDB RAG ← Vector store (auto-created)
api/ └─ chatbot_api.py	← FastAPI chat endpoints
start_system.sh requirements.txt	← One-command startup script

Complete requirements.txt

```
paho-mqtt==1.6.1
influxdb-client==1.36.1
kafka-python==2.0.2
transformers==4.38.0
peft==0.8.0
datasets==2.17.0
accelerate==0.27.0
bitsandbytes==0.42.0
torch==2.2.0
langchain==0.1.9
langchain-community==0.0.24
chromadb==0.4.22
sentence-transformers==2.3.1
ollama==0.1.7
fastapi==0.109.2
uvicorn==0.27.1
websockets==12.0
```

☐ Key Concepts Summary

Concept	What it does in YOUR system
MQTT Publish	Boiler/chimney Python scripts send sensor readings
MQTT Subscribe	InfluxDB consumer, Kafka bridge, fault detector all receive data
EMQX Broker	Routes messages, handles 25K+ connections, provides dashboard
QoS 1	Sensor readings delivered at least once (safe

Concept	What it does in YOUR system
	default)
QoS 2	Fault events delivered exactly once (critical data)
InfluxDB	Stores every reading with timestamp, enables time-range queries
Grafana	Live visual dashboard — no code, just queries
Kafka	Buffer that decouples fast MQTT from slower storage consumers
LoRA	Fine-tunes 7B LLM using 0.1% of parameters — GPU-efficient
QLoRA	4-bit quantized LoRA — fits 7B model on 16GB GPU
RAG	Injects real-time sensor data into every LLM prompt
ChromaDB	Stores embeddings of fault history for semantic retrieval
FastAPI	Serves the chatbot as REST + WebSocket API

Built with: Python · MQTT · EMQX · InfluxDB · Grafana · Kafka · Mistral 7B · LoRA · LangChain · ChromaDB · FastAPI · Docker